Seven big tips to keep your copy number-free and fair

In teaching reporters to use spreadsheets I've watched their eyes first widen, then glaze. Next comes frantic typing on the computer, hoping that a series of mouse clicks will compensate for an inability to focus on the job at hand. Reporters sometimes then realize that there are dozens of numbers on the screen and there's no right answer. The final step is an announcement to the rest of the class that, "I don't do numbers. They bore my readers."

It seems these reporters have given in to their fear of numbers. They'll always have difficulty reporting on a complex story if they can't overcome it. But how? Here are some simple tips to tame numbers and help you improve the copy that depends on them.

1. KEEP THE NUMBER OF DIGITS IN A PARAGRAPH BELOW EIGHT

Reducing the number of numbers in our copy can actually improve how we use the ones we're allocated. One suggestion, adapted from others, is to reduce the number of digits in a paragraph to eight.

Don't cheat by stacking paragraphs with eight digits on top of one another. Separate any number-packed paragraphs with anecdotes, sources' explanations, quotes or observations before you allow yourself more numbers.

This exercise, borrowed from William Blundell's "Art and Craft of Feature Writing," is geared toward forcing you to choose your numbers as carefully as you choose your quotes. It also forces you to find simpler ways to describe the numbers using words like "double" or "nearly" or "more than." And it forces you to demand that your sources simplify their numbers enough for you to summarize them.

Why eight? This usually gives you the ability to fit one year and two substantive numbers into a paragraph. Look for a way, though, to reduce this number of digits by simplifying to stress the most important point.

Here's an updated version of Blundell's example. You'd be over your allocation with a sentence like this:

The Office of Redundancy's budget rose 48 percent in 2001, from $700.3 million to $1.03 billion.

Think about how it could change:

Over the past year, the Office of Redundancy's budget grew by nearly half, to $1 billion.

If you follow this exercise, you'll find your copy reads more smoothly and conveys more meaning. You also may find yourself working more closely with the copy desk and your graphics department, using the space you save to convey information more effectively through graphics.

2. MEMORIZE SOME COMMON NUMBERS ON YOUR BEAT

The hardest part of dealing with numbers is answering this question: "Is it big or is it small?"

You can start to answer this question by memorizing some common numbers on your beat and keeping them in mind as you report.

Everyone needs to know the population of the U.S. (about 285 million at this writing), the size of the U.S. economy ($10 trillion right now) and the income of the typical household in the country (about $41,000 right now). These numbers are simply required if you expect to put perspective on any story with numbers.

Each beat has its own specialized numbers as well.

In education, they might include teacher salaries and the budget of the school district. In medical reporting, you might need to know the annual number of deaths each year in the U.S., the number of smokers or the leading causes of death. And in government reporting, the typical

campaign war chest and the local budget are crucial.

John Allen Paulos calls this "building a sense of scale." you can call it knowing the difference between a big number and a small number. If nothing else, it reduces errors in the newspaper.

For instance, one editor at a major newspaper changed a reporter's story over the weekend. Instead of calling, she simply changed the sales figure for a medium-sized retailer from something like $1.2 billion to $1.2 million. She later said she thought "billion" sounded too big. This was a case of failing to understand scale in business. If she'd known that $1 billion in sales wasn't particularly large for a retailer, or that Wal-Mart has sales of 100 times that amount each year, she'd have never made the mistake.

How can you memorize these numbers? It's not so hard, and you don't have to pull out an almanac and start memorizing the way you did in third grade. Instead, you can start by paying attention to the numbers you have to look up all the time.

For some, the easiest way to do this is to guess the number before looking it up. They're often wrong but it cements the correct number in their heads.

3. ROUND OFF – A LOT

Notice the numbers quoted so far: They're far from exact.

In fact, you can afford to be even less exact early in your reporting. In print or on the air, you'll want to make sure you know the most precise number before you characterize it.

But "characterize" is the important word here. Philip Meyer, in "New Precision Journalism," said it this way: "Decimal points are for meaning, not emphasis." Only use extra precision when it matters.

Early in your reporting, you can keep values in mind at an even more approximate level.

In fact, it would be perfectly respectable for you to know that the population in the U.S. is somewhere between 250 million and 300 million. That's plenty close enough to evaluate whether any other number is big or small. It also makes it easier to do some figures in your head.

This exercise reduces your dependence on exact numbers. It reduces the hair-splitting that many people find themselves in when dealing with numbers that don't deserve it. Sometimes you need an exact number. Sometimes you don't.

4. LEARN TO THINK IN RATIOS

Most of the math we use in the newsroom revolves around ratios of some kind: A percentage, a percentage difference, a rate or a value per person.

So cut most of the numbers you deal with down to size by learning to think in simple ratios. Don't try to think in terms of 1 million people. You can't picture it or write clearly about it. Instead, think in terms of 1 out of 285 people.

Let's take an example: The estimates of fixing the Year 2000 computer bug. A widely quoted estimate was $50 billion for U.S. companies. (Estimates for worldwide costs, usually cited in news stories, were about 10 times that level.)

How big is $50 billion? It's smaller than Bill Gates' net worth. It's the cost of two hurricanes. It's the income of people living in the Portland, Ore., area.

This isn't to say that the problem wasn't newsworthy. The possible widespread effect and the uncertainty surrounding it made the problem significant and newsworthy. But it wasn't a financial disaster.

5. USE DEVICES FROM EVERYDAY LIFE

Sports fans have a huge advantage when it comes to working with numbers. Basketball players think in terms of "1 for 10" or "2 for 8." Baseball fans already know how

to calculate a rate – either an earned run average or a batting average.

But even if you're not a sports fan, you usually have some things you're good at in math. Shoppers often can figure out percents in their heads because they know how much a $20 item ought to cost if it's offered during a 10 percent off sale.

If you shop on sale, invest in your retirement, follow sports closely or tip at a restaurant, you know how to do most of the math that we use in daily reporting. Simply convert your reporting questions into this scale.

For example, if you know that African Americans make up between 10 percent and 15 percent of the U.S. population, it doesn't take too much work to figure out how many people that is. It's like figuring a small tip on a restaurant bill.

ENVISION YOUR DREAM NUMBER AND CALCULATE IT IF IT ISN'T GIVEN TO YOU

Computer-assisted reporting trainer Drew Sullivan recommends that reporters "envision success" when struggling through one of the hardest parts of computer-assisted reporting. I've adopted this recommendation and ask reporters to look away from the keyboard, close their eyes, and picture the printout they'd get from a particularly helpful source.

This trick helps when you're away from the computer, too. Write your sentence, leaving out the key number. It helps if you have some clue what that number would be – 10 percent or 90 percent, $2 or $300,000. Now track down or calculate the number you need to fill in the holes.

You may find that the numbers don't support your preconceptions. Or you may find that the numbers you've been given so far don't answer any question you care about. That's frustrating. We all, at times, want to dump our notebooks onto the page. But it will help you figure out what's missing early enough to do something about it.

LEARN FROM ONE ANOTHER

Print and broadcast reporters have different strengths.

Broadcasters instinctively understand their viewers' inability to focus on too many numbers at once. A print reporter may disagree with the reporting approach, but these journalists haven't shirked their duty to simplify complex events. For print reporters, watching television news can help show how to select the key number for a story.

Print reporters often work well with their graphics departments to remove important numbers from copy and put them into charts. Broadcasters might consider the same technique. If nothing else, building graphics may help every reporter focus on the most important number for the story.

Good habits for reporting with numbers

Part of our fear of numbers comes from the good habits required to avoid embarrassing corrections and mispitched stories. These habits are boring. But if you get used to them, you'll often find yourself working more quickly at the end of a story.

USE THE TOOLS AVAILABLE TO YOU

There are tools to help us with arithmetic. Use them.

Anyone who works at a Windows workstation probably has a calculator right on the desktop. Many reporters keep a calculator (which I still find easier to use) in their desk drawers. Even old news production systems will do math for you.

If you learn a little about a spreadsheet, you'll find it's a particularly powerful calculator that repeats your formulas over and over with a single mouse click. That's good.

As Ronald Campbell, who teaches math to journalists at The Orange County Register says, you were never good at it anyway. So use the help that's available.

ALWAYS DOUBLE-CHECK YOUR ANSWERS

Whenever you do take out a calculator, be prepared to compute each answer at least twice. Not enough reporters do this.

You should do it. Always do it. And if you don't have time to do it, don't calculate any numbers.

For spreadsheet users, be prepared to manually spot check your formulas, double-check any numbers you type, and confirm simple summary measures with any totals you can find and the raw numbers from a printed report.

Some reporters develop particularly good habits. They print out a draft of the story, then circle every number (and every name) in it and double-check it. This includes years, numbers given to you by a source and numbers you calculate yourself. Try it. It might frighten you into good habits when you discover how often mistakes are buried in your copy.

Although it's not always possible, it helps if you can work with another reporter, a copy editor or someone else you trust during the math phase of your reporting.

ASK YOURSELF IF THE ANSWER MAKES SENSE

Always force a gut-check on your numbers. Use them in a couple of sentences. Ask yourself whether you'd believe them if a fellow journalist (rather than a math-savvy expert) gave them to you.

If you can't figure out if you believe them, then it's time to go back to some of the other tips in this book – working out ratios, rounding off, or looking up some general numbers on the topic.

In a letter to The Washington Post in October 1998, a reader challenged an op-ed piece describing the Census Bureau's plans to estimate information for 27 million people in 78,000 households. The letter writer surmised something was wrong with the published numbers. The reader was right. Such an estimate would lead to almost 350 people in each of the affected households.

Some numbers, like this one, simply don't stand up to common sense. If yours don't, make sure you haven't misunderstood the calculation, mistyped the numbers, or misread the results.

ROUND – DON'T CHOP OFF DIGITS

Make sure you round numbers rather than chop off digits when you want to simplify your answers. If the next digit is 4 or smaller, you round down. If the next digit is 5 or bigger, round up.

So, 342.5 rounds to 343, but 342.4 rounds to 342.

USE PRECISE LANGUAGE – IT'S NO MORE COMPLEX OR DIFFICULT THAN IMPRECISE LANGUAGE

Don't confuse "percent" with "percentage points," or a raw change with a percentage rate. It usually takes no more words to say your answer accurately rather than inaccurately. But failing to accurately characterize the numbers you use stands along with misspelled names and errors in factual details that kill your credibility with readers, viewers, sources and even editors.